

Contextualizing Intermediated Use in the Developing World: Findings from India & Ghana

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ABSTRACT

This short paper extends the existing conceptualization of intermediated use in the developing world by demonstrating a range of informal practices that are conducted outside of a discrete (intermediary/beneficiary) user-interface interaction in a given point in time. Further, this paper also demonstrates how low-literate users may often voluntarily relinquish custody of an information resource in order to create and maintain intermediation. In this way we describe a broader conceptualization of intermediated use in the developing world that needs to take into account the entire sociotechnical workflow. This is particularly critical when explicitly designing for secondary/beneficiary users; it considers their specific requirements that often get overlooked while simultaneously revealing their vulnerabilities within the formal-informal continuum. We present findings from ethnographic work conducted in India and Ghana.

Author Keywords

Intermediated Use; Informal Practices; HCI4D

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

INTRODUCTION

It is well-understood by now that the basic tenets of HCI design, having originated in the developed world, is not conducive to a direct transfer to the developing world if at all one wants to build usable and relevant systems [1]. Of particular interest to this paper is the heavy presence of intermediated interactions in the developing world, as has been studied by Parikh & Ghosh [2] and Sambasivan et al [3] previously. Parikh & Ghosh describe how a proxy primary user (that is the “intermediary user” as Sambasivan

and her co-authors say) may manage the information-seeking experience (most likely, but not necessarily, through a technology platform) of a secondary user (or “beneficiary user”), because they possess the skills to do so. Often, in the developing world, particularly in low-resource settings, these skills translate to digital and textual literacy. The beneficiary user’s use of the information resource at hand, therefore, becomes “intermediated”. The authors also make the valuable distinction that is when an intermediated task is completely abstracted from the beneficiary user, it then becomes an “indirect” interaction. Sambasivan et al, refocus their attention on technology devices as information resources, and remain careful in distinguishing such intermediation from instances of spot assistance by demonstrating how such informal help “fundamentally” enables technology access and use for a large part of the population(s) across the developing world. The authors further differentiate between different types of interaction mechanisms: i) “surrogate usage” that sees the intermediary user both inputting as well as interpreting the output for the beneficiary user ii) “proximate enabling” that sees the intermediary user inputting the requisite commands thereby simplifying the device output for the beneficiary users, and finally iii) proximate translation that sees the intermediary user interpreting the device output for the beneficiary users [3]. Still, the discussion of intermediated interactions in the HCI literature has thus far largely confined itself to that specific point in time when the intermediary and beneficiary users are in some way directly interacting with the technology/information resource. In other words, these intermediated interactions are discrete, time-bound events that collapse, theoretically, if the technology device or the information resources is removed from the picture. Sambasivan et al [3] do account for remote intermediation in their description of surrogate usage. However, such intermediation is still very much bound by its dependency to some form of interaction with a technology or information resource, without any explicit attention paid to the broader socio-technical workflow and the host of informal actions that take place outside of that discrete interaction. Furthermore, they do not account for the voluntary surrender of custody in their categorization of different factors that motivate intermediation, specifically when users are in clear ownership of an information resource or digital device. Thus, intermediated use within

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the broader socio-technical system becomes even more relevant when indirect intermediated interactions are considered – the secondary/beneficiary user receives no process feedback and often, no immediate output feedback either. It is useful to write them into these taxonomies of intermediated tasks to ensure that they are represented, at least for analytical purposes, even though they are otherwise invisible in these processes. But perhaps even more importantly, it is useful to demonstrate the scope of informal practices within a formal socio-technical system that not only reveal their specific needs but also ensure that the beneficiary user receives salient information during the process to increase their participation, trust, and awareness in the overall sociotechnical workflow.

Such intermediated actions align well with informal practices and processes that abound in the developing world. In general, the formal-informal sector dichotomy has prevailed in the development discourse for a while now, although it has failed to enter the HCI for development discourse, especially in terms of the design implications it bears for the field at large. Scholars studying this over the years have now established that such a dichotomy may in fact be spurious; instead, they contend, there is a formal-informal continuum in practice [4]. Nowhere is this formal-informal continuum more visible than in low-resource settings in the developing world where low-income, typically rural residents, may often find themselves at a significant distance, both physically and culturally, from the state. The state may therefore be unable to fulfill their specific utility needs which gives rise to informal practices that can generally address these deficiencies in a timely and salient manner. Such informal practices operate outside the precincts of state rules and regulations, although they are not immediately considered as illegal (Chen [5] makes the valuable point that there must be a differentiation between illegal “process” and “product”); instead they may be a direct response to very restrictive official regulations or, as mentioned previously, to where such regulations are unable to address the needs of specific populations. Eventually, we contextualize indirect, intermediated actions within the larger socio-technical workflow and demonstrate the scope of informal practices used to achieve this, in this short paper. This reveals the specific requirements of low-income, low-literate populations while telling, simultaneously, their particular vulnerabilities when interacting with technical systems or with digital tools. More specifically, we present two case studies in the domains of mobile banking and microfinance from India and Ghana; domains that are, ostensibly, well structured and regulated, and thus formalized. Of these case studies, one considers the intermediated use of a technology application and another that considers the intermediated use of a paper passbook, which is not a technology device but constitutes a valid information resource that is challenging to parse for low-literate individuals. Therefore, there are

lessons to be learnt from this, as even Parikh & Ghosh have demonstrated previously [2].

METHODOLOGY

We present findings from two separate ethnographic studies conducted in India and Ghana in the years 2012 and 2014 respectively. Fieldwork was undertaken in New Delhi, India over a period of two and a half months between June and August 2012, whereas fieldwork in Tema, Ghana was conducted over three weeks in July 2014. Respondents in both field sites were low-income and typically low-literate¹. Specific methods included a combination of in-depth interviews and observations in both cases. It must be noted that the author is familiar with Hindi that was the dominant language spoken in New Delhi. The author used the services of an interpreter in Ghana where all interviews were conducted in either Ga or Twi. All field notes and interview transcripts/notes were subjected to a thematic coding process; indeed the data from both countries was revisited for another round of coding and analysis for this paper.

MOBILE BANKING IN INDIA: INTERMEDIATION OUTSIDE OF INTERFACE-INTERACTION

The branchless banking model in India sees formal financial institutions form a hybrid partnership with local organizations (only certain types of organizations have been enabled by regulatory body to enter such partnerships) to extend financial services to rural areas and/or low-income populations. Traditionally, such populations have been expensive to service, but hybrid arrangements such as these can mitigate the costs as well as the risks involved by enabling remote transactions outside of a brick and mortar bank branch. Technology Service Providers, or TSPs are they are more concisely called, have proven to be one type of local organization that provides both the technology platform that supports these remote transactions, as well as the retail agent network that can enable the transactions to complete the digital-liquid-digital currency exchange.

Medhi et al [8] in their study on mobile banking agents in India, Kenya, South Africa, and the Philippines find that low levels of textual and digital literacy compounded with instructions being in a non-native or local language, and a lack of familiarity with banking jargon led customers to heavily rely on their agents for any transactions that were to be conducted via a mobile platform. As we have previously reported in [6], we find this in our fieldwork as well, further coupled with a fear of completing a transaction incorrectly. In general, our respondents unanimously trusted their more “skilled” agents with this task. In fact, completing the transaction on your own is less expensive than relying on an agent to complete it. Still, we found not one customer, nor had the agents encountered anyone, who was merely

¹ Please refer to [6] and [7] for a more elaborate description of our sampling methods and the details of our respondents.

topping his/her account with digital currency to later conduct the transaction on their own.

Parihk & Ghosh [2] observe that providing “process feedback” during an intermediated interaction, especially if it is an indirect interaction, will improve the beneficiary users’ awareness and trust in the overall workflow. We also observed that agents will often hand out a paper slip with relevant information, such as bank account details, the amount remitted, the fee charged, the transaction ID number, and the agent’s personal number in case of irregularities, as a provisional paper receipt to their customers [6]. This is done despite the receipt of SMS receipts directly from the TSP on the customers’ mobile phones. In other cases, agents would steal deposit slips from the physical branches of the banks they were affiliated with. Agents would have customers fill this out, and then stamp and sign it, therefore simulating a familiar banking process that is routine in brick-and-mortar bank branches [6]. These events corroborate Parikh & Ghosh’s assertions about paper being a time-honored interface between intermediary and beneficiary users, especially in the domain of microfinance. But more importantly, they describe the informal practices within a workflow that is largely intermediated, and frequently indirect where the intermediary conducts the transactions from behind his wooden counter or table thereby blocking even visual access for the beneficiary user. These informal practices on the part of the intermediary, that is the agent, communicates some salient information about the process to the beneficiary users, such as which account money was remitted to or deposited in, and how much money it was, and so on. Such process feedback, while mainly abstracted from the actual technical procedures that the intermediary engages in, continues to reinforce the beneficiary users’ trust in a technical system that remains largely inaccessible to them. It is also worthy of note that often such informal practices may in fact be modeled on existing formal (and familiar) processes (such as the completing of and then receipt of a signed and stamped deposit slip) to ensure that feedback is particularly meaningful.

MICROFINANCE IN GHANA: INTERMEDIATION WHEN CUSTODY IS VOLUNTARILY RELINQUISHED

We now describe a microfinance institution in the city of Tema in Ghana that brings financial services to the doorstep of low-income, typically unbanked populations in their immediate vicinity. This MFI has been previously studied in [7]. Essentially, the MFI sends out mobile bankers, or susu (that is, savings) collectors as they are called, to collect the savings deposits from their customers directly. Customers will choose a standard deposit amount that they think they will be able to put aside on a daily basis. They then save this amount with the mobile banker over a 31-day cycle. At the end of 31 days, the MFI makes available 30 days worth of deposits to the customer for withdrawal, with

the deposit on the 31st day going towards their service fees. This information is maintained in a paper passbook, an information resource that Parikh & Ghosh dedicates some attention to in their paper when discussing intermediated interactions [2]. This paper passbook is not a technology device (although some would argue otherwise) but it constitutes a legitimate information resource that, depending on its esoteric content, can still require low-literate users to borrow skills from a proxy primary user leading to intermediated use. As we saw during the course of our fieldwork, the susu collector would interact directly with the passbook to update it. Still, the passbook was expected to remain with the customer once a transaction was complete to ensure that they were aware of their balance at all times. However, we found that many customers asked their susu collectors to safeguard their passbooks for them. This was because they were afraid of getting them wet or soiled in the open markets where many of them were working. They recognized the passbook’s value and were confident that it would be safer with the susu collector than with them. Furthermore, as we describe in [7], customers were able to trust the susu collectors through a combination of interpersonal trust, where the susu collectors themselves worked hard to develop and maintain this, and institutional trust, where the longevity and reputation of the MFI and its affiliated organizations, continued to reinforce the customers’ trust in the overall proceedings.

Interestingly, customers kept a running tally of their increasing balance in their heads. This process could get particularly complex since customers were not always able to make their deposit payments every single day. Therefore, to make up, they would deposit a larger amount on a later day, essentially leading to variable deposit amounts over time. This complicated the math considerably. However, customers were able to reproduce the immediacy of determining their balance from a passbook (which they could not always accomplish when actually reading the passbook) by using their mobile phones to make a quick phone call to their susu collectors, especially before they intended to make a withdrawal. In such cases, they would inquire about their current balance and then make a spot decision about how much money they wanted to withdraw and thus inform the susu collector accordingly. In other cases, they would leave the withdrawal amount up to the susu collector’s discretion. As one customer told his susu collector on hearing his balance, “leave some, and bring some tomorrow.” Again, it is important to note that the prevalence of informal practices (that is the safeguarding of the passbooks by the susu collectors which, in turn, prompts phone calls to determine exact account balance) continues to provide salient information to the beneficiary users of the passbook. In this case, this information is not so much about the process itself, but about the output that needs translation.

DISCUSSION & CONCLUSION

Intermediated Use & Informal Practices

We present findings from ethnographic fieldwork in India and Ghana that demonstrate the range of informal practices that are used to directly subvert the rubrics of regulated, structured, and formalized financial sectors in both countries. Elsewhere, research has shown that the presence of such informal practices, typically on the part of the intermediary user, to better enable and support the beneficiary users can successfully determine the adoption and uptake of a service [8]. This is especially important when considering settings where low textual and digital literacy is prevalent, and people rely heavily on familiar markers of trust, especially those derived from the formal sector, to even try out a new service. For instance, the stamping and signing of deposit slips that were stolen from bank branches were used to both reassure customers as well as communicate process feedback to them. When this was not feasible, even regular paper slips or receipts served the purpose. These bore no actual legitimacy, of course, but were preferred to the standard SMS receipts that customers were receiving instead on their mobile phones and that they could not parse for the most part. Similarly, customers voluntarily surrendered custody of their passbooks by trusting their susu collectors to safeguard them, and leveraging the immediacy that the mobile phone provides to have persistent access, at least in theory, to their account information. Of course, users are less likely to give up possession of their digital devices because they are more expensive and support a wider range of functionalities; still it remains critical to codify them as informal practices continue to emerge unpredictably in response to very specific needs. Eventually, the agents and susu collectors, on their part, facilitated intermediated use that went beyond the actual interface-interaction, be it in stealing deposit slips in their spare time or fielding calls from customers throughout the day. Such intermediated use within the larger sociotechnical system, and not just at the point of service, addresses and reveals the specific needs of the target population that is overlooked by the application design, and communicates salient information to them about the process or the output, or both, which further improves their participation in the proceedings, if even marginally. Therefore, it is absolutely imperative that we think about intermediated use in this broader sense.

However, at the same time, it is absolutely critical to note that these informal practices expose loopholes in the systems that can be easily exploited. These seemingly institutionalized formats that low-literate users trust and rely on can be just as easily manipulated to engage in instances of fraud, another aspect of intermediated interactions that has not been discussed in the HCI literature thus far. While such informal networks/practices generally thrive and work to address gaps in the socio-technical system (in fact, we did not directly observe any instances of

fraud; we only encountered them through hearsay), documenting such practices from the field can be helpful in including some checks and balances along the way that may account for potential fraud. For instance, the MFI in Tema always calls customers from the office to confirm withdrawal requests that susu collectors make on their behalf. We absolutely do not recommend a wholesale elimination of such informal practices – eventually intermediated use/interaction is also a type of informal practice (although not always) that helps low-literate users gain access to and navigate socio-technical systems.

Design Considerations

Finally, we wish to conclude by revisiting some of the design recommendations that Parikh & Ghosh [2] and Sambasivan et al [3] make in their papers. Both papers recommend that future design endeavors must explicitly include the perspectives and address the requirements of secondary/beneficiary users. We agree with this, even though existing designs continue to assume individual use case scenarios, or focus on the intermediary-computer interactions. We believe that taking a step back to evaluate the larger picture, that is beyond the interaction at the point of service, will actually help shed light on the needs of the beneficiary users (their reliance on paper receipts for instance, or their inability to always secure their passbooks). Such detail may or may not be revealed in the needs assessment phase; often they may be discovered retroactively when existing technology applications or information resources are already in use. This is because informal practices emerge spontaneously in response to gaps in the socio-technical system, and it is these practices that can be profoundly telling of these gaps.

We also agree with their recommendation that future designs endeavors should aim to mitigate what Sambasivan et al refer to as the “absent presence” of beneficiary users by involving them more in the user experience. Parikh & Ghosh call for more process feedback and Sambasivan et al call for more digital habituation, participation, and transparency. Again, it is helpful to consider the larger socio-technical system and learn from the informal ways in which intermediaries are already trying to involve beneficiaries. It will be especially helpful if the primary/intermediary user is able to include technical process feedback into their informal apparatus as well; for instance, mobile agents may want to work through the digital receipts with their customers, that is actually legitimate as opposed to the paper receipts. Letting them observe the transactions more directly might also help them familiarize and learn the procedures over time, and mitigate the beneficiary users’ absent-presence status to a greater degree.

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